BA820 – Project Proposal

Cover Page

- **Project Title:** Exploratory Analysis of Structural Patterns in Complex Datasets
- **Section and Team Number:** B1 Team 10
- **Members:** Parisha Patel, Inchara Ashok, Shanmathi Shivkumar, Bhavya Bavishi

1. Primary Dataset

1.1. Project Motivation and Preliminary Exploratory Data Analysis (EDA)

Hollywood films shape cultural ideas about romance, aging, and gendered power. Age disparities in on-screen romantic pairings, especially between older men and younger women, are often criticized, but empirical evidence is limited. This project uses the Hollywood Age Gap database, covering 1,155 couples in 830 films since 1935, to examine structural patterns in age gaps. Understanding these patterns informs both cultural analysis and industry practice, as casting norms influence audience perceptions and career trajectories. In the post-#MeToo era, studios face pressure for greater transparency in representation. Stakeholders include film studios, casting directors, diversity and equity organizations, researchers, and audiences interested in media literacy.

Preliminary Exploratory Data Analysis (EDA)

Age gaps vary substantially across decades and do not follow a smooth temporal trend (Fig. 1). Early decades exhibit extremely large median gaps, followed by sharp mid-century declines and more modest variation thereafter. These changes appear as discrete inflection points rather than gradual shifts, and the data is heavily imbalanced across decades, motivating a focus on decade-level comparisons rather than overall averages.

At the director level, pronounced heterogeneity emerges (Fig. 2). Directors cluster into distinct groups, with a small number of extreme cases, suggesting that age disparities may reflect persistent creative tendencies rather than project-specific noise. Within films that feature multiple romantic couples, age gaps are unevenly distributed (Fig. 3). Some films concentrate disparity in a single dominant pairing, while others display more consistent gaps across couples, motivating questions about whether age disparities function as isolated narrative choices or broader casting patterns.

Finally, comparisons across couple types reveal strong representational imbalance (Fig. 4). Mixed-gender pairings dominate the dataset, while same-gender couples are sparse and highly variable. Despite this instability, visual summaries suggest differences in typical age gaps, motivating comparative questions that explicitly account for uncertainty.

An unexpected outcome would be finding little variation in age gaps across directors, decades, or gender compositions, suggesting that industry-wide pressures, rather than individual choices or historical shifts, drive casting patterns.

1.2. Domain / Business Questions

Preliminary exploratory data analysis (EDA) indicates substantial variation in on-screen romantic age gaps across time periods, filmmakers, and couple types, suggesting that casting age disparities are not random but structured by industry norms and creative decision-making.

1. **How has the "typical" age gap in on-screen romantic couples shifted across decades (1930s–2020s), and where do the most significant inflection points occur? -** This question examines whether changes in age-gap norms align with broader social and industrial shifts, such as the studio system era, New Hollywood, or the rise of streaming platforms.
2. **Do directors cluster into distinct "casting age-gap profiles" (e.g., consistently large gaps versus age-balanced casting), and how stable are these profiles across their careers? -** This explores whether age disparities are driven by persistent creative patterns at the director level rather than isolated project-level choices.
3. **In films featuring multiple romantic couples, do age gaps concentrate in a single "headline" pairing, or do they systematically recur across couples within the same film? -** By analyzing within-film variance, this question investigates whether age gaps function as deliberate narrative devices or reflect broader casting norms embedded in individual productions.
4. **How do age-gap distributions differ across couple types (mixed-gender, man–man, woman–woman), and are same-gender pairings closer to age parity? -** This comparison helps disentangle whether age inequality in romantic casting is primarily gendered or reflects wider industry economics and star power dynamics.

Stakeholders who would benefit from these insights include film studios, casting directors, creative executives, and diversity, equity, and inclusion (DEI) teams. Understanding long-term shifts in age-gap norms allows studios to evaluate whether their casting practices are evolving in step with audience expectations or lagging behind social change. Identifying director-level age-gap profiles supports more informed greenlighting, hiring, and accountability decisions by highlighting systematic creative patterns. Analysis of within-film age-gap concentration clarifies whether disparities are intentional storytelling choices or indicative of structural casting practices. Finally, comparing age gaps across couple types informs representation strategies, risk assessments related to audience perception, and broader discussions around gender and age equity in media.

2. Backup Dataset

2.1 Preliminary Exploratory Data Analysis (EDA)

The goal of our exploratory data analysis (EDA) was to identify structural patterns, imbalances, and sources of variability in the fair use dataset to motivate substantive unsupervised research questions rather than produce final results. At the case level, disputes frequently span multiple content categories, while a small number of media types—particularly textual works, film/audiovisual content, and photographs—account for a disproportionate share of cases (Fig. 5 and 6), suggesting recurring combinations of content type and use context and motivating our first question on latent case archetypes.

The dataset also shows substantial imbalance across courts and jurisdictions, with a few circuits contributing most cases and notable variation in fair use outcomes (Fig. 7). Differences in average holding length across courts indicate variation in how legal reasoning is structured (Fig. 8), motivating our second question on jurisdiction-level reasoning patterns. Textual EDA further revealed that issue statements are short and tightly distributed, while holdings are much longer and highly variable (Fig.9), suggesting that similar framing can mask meaningful differences in reasoning and motivating our third question on semantic similarity across holdings. Finally, transformative-use language appears frequently but is applied inconsistently across opinions (Fig. 10), motivating our fourth question on recurring reasoning patterns in transformative-use analysis. An unexpected outcome would be that courts and jurisdictions show little difference in reasoning patterns once case facts are accounted for, suggesting greater doctrinal uniformity than typically assumed.

2.2. Domain / Business Questions

Preliminary exploration of the legal text corpus suggests that fair use disputes exhibit recurring but implicitly structured patterns in factual contexts, judicial reasoning, and interpretive language.

1. **What latent "case archetypes" emerge across fair use disputes based on factual context and judicial outcomes?** - By clustering cases using semantic embeddings derived from key facts, legal issues, and holdings, this question seeks to surface dominant dispute archetypes without imposing predefined categories.
2. **How do courts or jurisdictions differ in their underlying reasoning patterns when deciding fair use cases, absent prior labeling?** - This question examines whether distinct "reasoning styles" emerge by clustering judicial holdings and comparing the distribution of resulting clusters across courts and jurisdictions.
3. **Can precedent similarity be operationalized to support decision-making for new creator use-cases?** - Using semantic nearest-neighbor retrieval over embedded case texts, this question explores whether new hypothetical scenarios can be meaningfully mapped to past cases and associated archetypal reasoning "playbooks."
4. **What recurring language and logical patterns characterize findings of "transformative use," and how do these patterns group?**- By clustering holding text alone, this analysis aims to extract reusable argument templates, common phrasing and logical structures, that courts employ when evaluating transformative use.

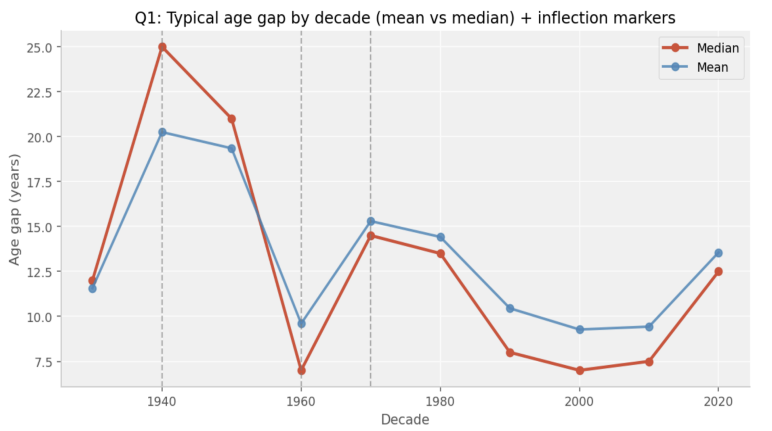
These insights support creators, legal practitioners, platform policy teams, compliance officers, and product teams. Identifying case archetypes and jurisdictional reasoning patterns enables faster risk assessment, better policy design, and more consistent guidance for fair use decisions. Distilling reusable "transformative use" argument templates further supports legal education and assistive drafting tools.

3. Appendix (Optional but Highly Recommended)

GitHub Repository (Required) - https://github.com/patelparisha/ba820-b1-team10

Supplementary Material (Highly Recommended)

Figure 1:



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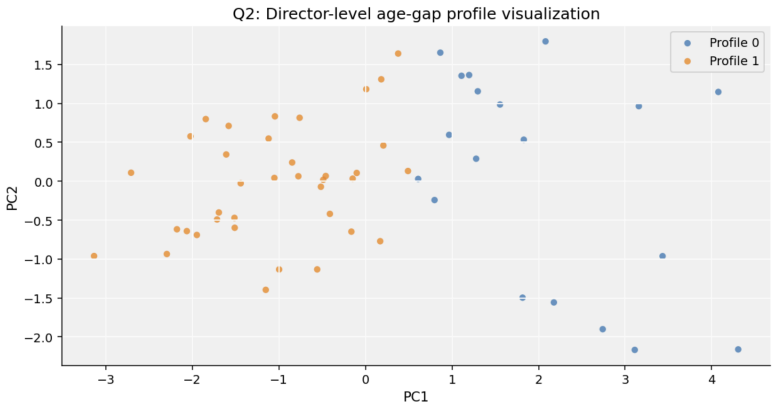

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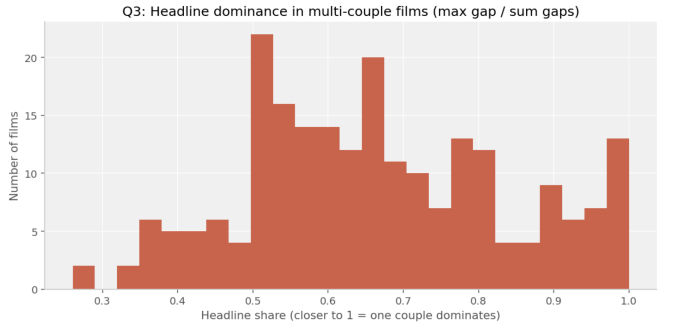

Figure 4:

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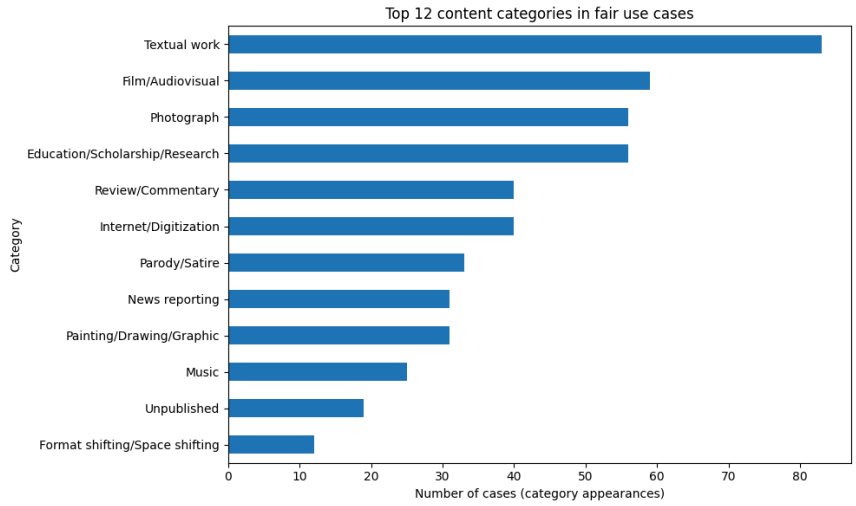

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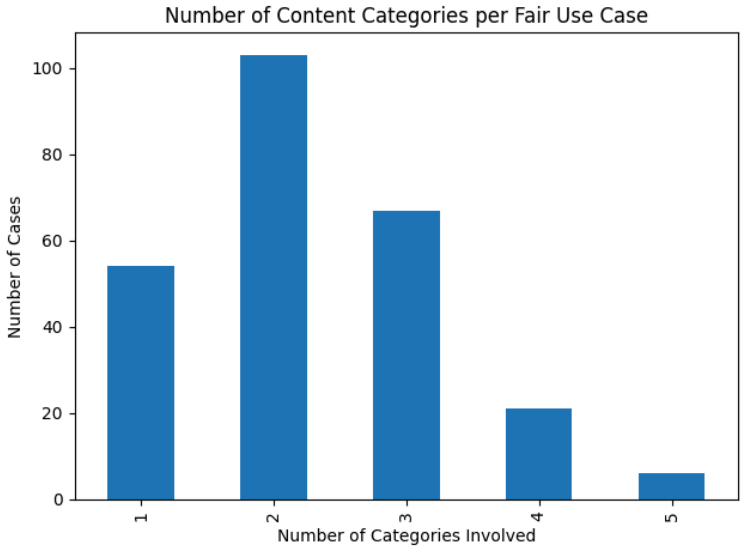

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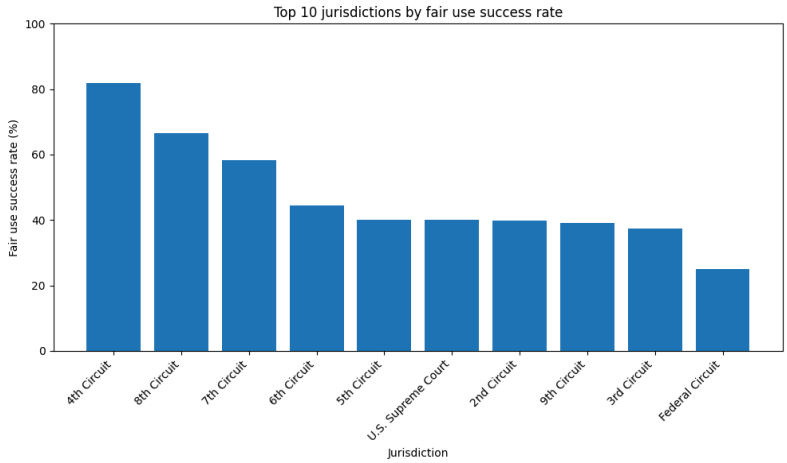

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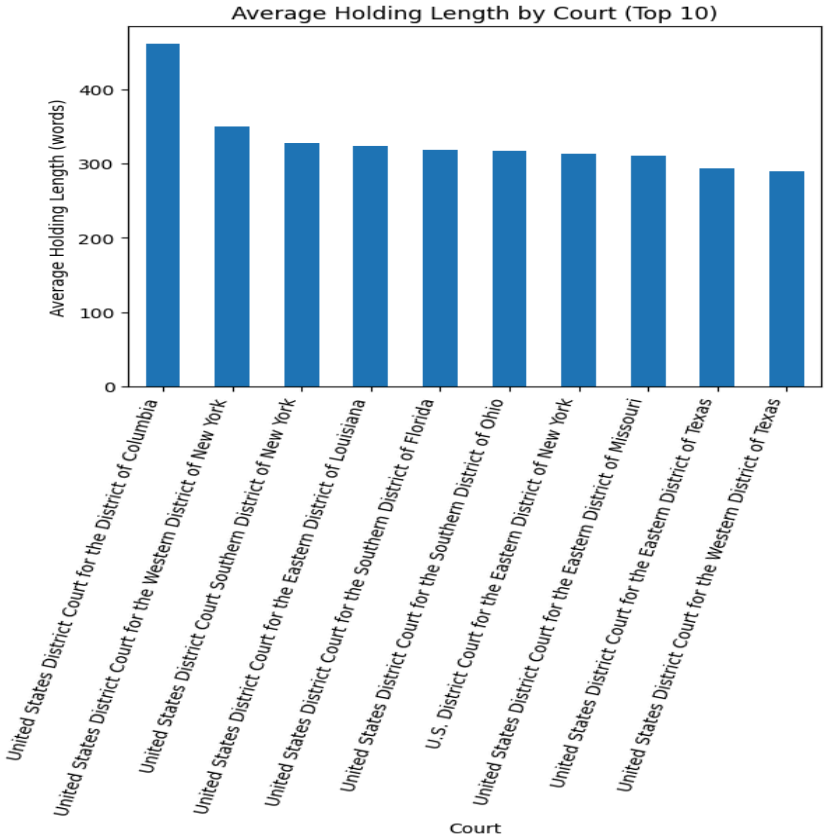

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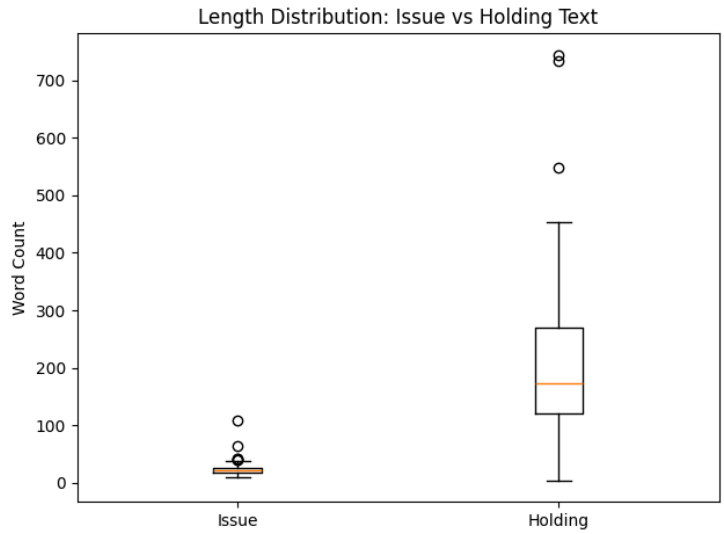

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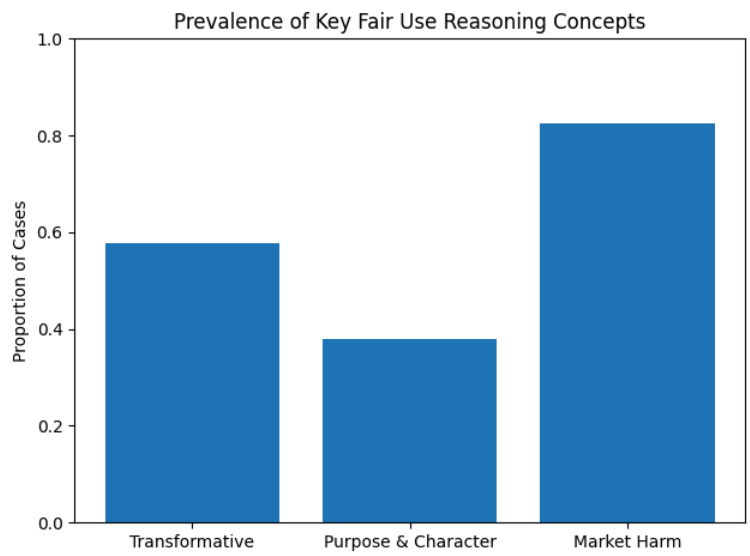

Process Overview

The project began with the selection of two datasets from a curated list provided by the instructor, with the Hollywood Age Gap dataset chosen as the primary focus and the Fair Use dataset retained as a secondary option. As both datasets were pre-cleaned, the initial phase centered on understanding their structure, scope, and domain relevance rather than performing data cleaning. The team then established the project motivation, identified key stakeholders, and articulated how each dataset could be used to examine meaningful real-world questions.

Exploratory data analysis was subsequently conducted to examine distributions, temporal trends, and structural characteristics within each dataset. Insights from this exploratory phase directly informed the formulation of domain-specific questions, resulting in four guiding analytical questions for each dataset. These questions were designed to support the discovery of latent patterns and recurring structures without imposing predefined labels. Finally, the process set the foundation for deeper pattern exploration and interpretation in later stages of the project.

External References and Context

1. Project Fabulous Evee. (2023). *Final project report* (INFO 2950, Spring 2023). Cornell University. https://pages.github.coecis.cornell.edu/info2950-s23/project-fabulous-evee/report.pd
2. Bishop, Katie. "Age Gaps: The Relationship Taboo That Won't Die," March 24, 2022. https://www.bbc.com/worklife/article/20220317-age-gaps-the-relationship-taboo-that-wont-die.
3. "An Empirical Study of U.S. Copyright Fair Use Opinions Updated, 1978-2019 – NYU Journal of Intellectual Property & Entertainment Law," n.d. https://jipel.law.nyu.edu/an-empirical-study-of-u-s-copyright-fair-use-opinions-updated-1978-2019/.
4. Madison, Michael J. "A Pattern-Oriented Approach to Fair Use." *William & Mary Law Review* 45, no. 4 (2004): 1525–1688. https://scholarship.law.wm.edu/wmlr/vol45/iss4/5.
5. Crews, Kenneth D. "Decoding Academic Fair Use: Transformative Use and the Fair Use Doctrine in Scholarship." *Journal of Copyright in Education and Librarianship* 1, no. 1 (2016). https://journals.ku.edu/jcel/article/view/6481.
6. Wang, Yifan, et al. "Dissecting Judicial Reasoning in U.S. Copyright Damage Awards." *arXiv* (2026). https://arxiv.org/abs/2601.09459.

Contributions

Parisha Patel coordinated the project workflow and overall structure of the proposal, contributed to dataset selection, conducted exploratory data analysis, and helped synthesize findings into domain-driven questions. Bhavya Bavishi focused on evaluating dataset feasibility and structure, performed exploratory visualizations and descriptive analysis, and contributed to identifying meaningful patterns and irregularities. Inchara Ashok carried out exploratory data analysis with an emphasis on trends, variability, and relationships in the data, and assisted in interpreting EDA results to shape insight-oriented questions. Shanmathi Shivkumar translated analytical insights into clear domain narratives, contributed to the development and justification of business questions, and supported drafting and editing to ensure coherence and alignment with project objectives.

Use of Generative AI Tools

We used ChatGPT as a supportive tool to help with idea generation, drafting text, and clarifying

technical or conceptual questions during the project. All suggestions provided by the AI were critically evaluated, refined, and adapted to fit our own approach and understanding. The final work represents our own analysis, reasoning, and contributions, and AI was used only as an assistant, not as a replacement for our thinking.

Documentation of AI Usage:

1. https://chatgpt.com/share/697a99e5-8658-8010-b984-926696ed4c16
2. https://chatgpt.com/share/697aab38-c49c-800f-a175-8ffcba9fac60